

University-integrated health territories to tackle rural health-care inequities in France



Evidence suggests that recruiting medical students from rural backgrounds and training them in underserved areas increases long-term retention in the health-care system,^{1,2} and international examples substantiate this notion. In Australia, rural clinical schools require a full year of rural immersion, which triples the likelihood of rural settlement.^{3,4} In Canada, more than half of Northern Ontario School of Medicine graduates work in remote regions.⁵ In the USA, the Washington, Wyoming, Alaska, Montana, and Idaho (WWAMI) programme maintains over 50% retention of physicians in underserved areas.⁶

In France, the Sarthe and Mayenne departments remain among the most underserved, with an accessibility index for general practitioners 30% below the national average.⁷ By contrast, Angers, which hosts the university hospital and the health faculty serving the entire region, concentrates most academic and clinical resources. Inspired by international models of rural medical education, the Territorial University of Health Sarthe-Anjou-Mayenne (TUS-SAM) was launched in 2019 to apply these principles, including early exposure of medical students to rural clinical practice, integration of interprofessional training, and structured support for students in underserved areas. This initiative combines educational decentralisation, local capacity building, and community partnerships to strengthen the rural health workforce.

In addition to Angers, first-year health sciences courses are now offered in Laval and Le Mans, two towns that previously lacked medical training. This expansion allows students to study closer to home and promotes early exposure to rural health care. Decentralised teaching includes simulation centres, interprofessional learning, and standardised assessments, ensuring quality across sites. Placement opportunities outside Angers grew between 2019 and 2023 from 54 to 78 in Mayenne and from 198 to 263 in Sarthe. By 2023, 18% of second-cycle placements from the health faculty, which covers the Sarthe, Maine-et-Loire, and Mayenne departments, took place in community hospitals. A 2024 survey found that 85% of students were highly satisfied with these rural placements, emphasising the benefits of close mentorship and autonomy (unpublished). 47% of the respondents expressed an intention to practice in rural or semirural

areas, with results similar to those from WWAMI and Northern Ontario.⁵

TUS-SAM has created academic positions for early-career and senior physicians in community hospitals. These clinical fellowships combine teaching, research, and clinical care, allowing young doctors to build academic experience while practicing locally. Since 2020, 23 positions have been co-funded by local authorities and health agencies, strengthening education and local expertise through new or expanded services in neurology, nephrology, and geriatrics.

Retention results are encouraging, with 12 of 13 physicians completing a TUS-SAM fellowship staying in the region. The TUS-SAM fellowship complements shared hospital-university fellowships, in which practitioners divide their time between the university hospital and community sites, achieving a 79% retention rate since 2009 in the faculty coverage area (unpublished). This dual strategy of training students locally and embedding academics in community hospitals has created a coherent pipeline consistent with international rural education models while adapting to French governance and funding.

TUS-SAM's shared governance unites the university, hospitals, and local authorities—a distinctive model rarely described internationally. The initiative also decentralises advanced practice nurse training, supported by simulation-based teaching and interprofessional assessments that ensure consistency. This approach strengthens collaboration and resilience in rural health care.

Within 4 years, the TUS-SAM programme has produced measurable results. The proportion of second-year medical students from Sarthe and Mayenne increased from 30.9% to 40.9% (unpublished), demonstrating that local training improves recruitment and retention. A Regional Clinical Research Directorate and a primary care research network now connect general practitioners and hospital physicians across the territory. Since 2021, more than 40 decentralised projects have been launched—twice as many as in the previous decade. Community hospitals now recruit more than 1000 patients annually into clinical studies, compared with fewer than 500 patients before the introduction of TUS-SAM. These initiatives have given community hospitals a clear academic identity and



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measurable impact. Education, research, and innovation are now aligned through regional partnerships, enabling multisite trials and prevention projects.

Logistical barriers, mainly transport and housing, remain challenges, although partly offset by local government co-funding. Similar issues are reported internationally, highlighting the need for sustained political and institutional support.⁵

Sustainability depends on a co-funding mechanism linking the University of Angers, regional hospitals, the Regional Health Agency, and local authorities. Together, these entities support academic positions, infrastructure, and mobility. Long-term stability requires embedding this model in national frameworks, as in Australia and Canada, with federal mechanisms ensuring lasting support for rural education. For policy makers, the message is clear: decentralising academic health functions is not merely an educational reform but a strategic investment in territorial equity, workforce sustainability, and citizen trust. The TUS-SAM model shows that integrating education, research, and care at the regional level can transform underserved regions into resilient health ecosystems.

We declare no competing interests.

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*Mathieu Levailant, Cécile Jaglin-Grimonprez, Christophe Verny, Nicolas Lerolle, Cédric Annweiler
mathieu.levailant@univ-angers.fr

Faculté de Santé, Département de Médecine, Université Angers, Angers 49045, France (ML, CA); Centre National de la Recherche Scientifique, Inserm, Arènes, UMR 6051, Recherche sur les Services et Management en Santé, Université de Rennes, Rennes, France (ML); General Direction, Angers University Hospital, Angers, France (CJ-G); Neurology Department, Angers University Hospital, Angers, France (CV); Institut MitoVasc, Structure Fédérative de Recherche Interactions Cellulaires et Applications Thérapeutiques, Centre National de la Recherche Scientifique, UMR 6015, Inserm U1083, Université Angers, Angers, France (CV); Department of Anesthesia and Critical Care, Angers University Hospital, Angers, France (NL); Department of Geriatric Medicine and Memory Clinic, Research Center on Autonomy and Longevity, Angers University Hospital, Angers, France (CA)

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